Rahul Dange

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Sydney, Australia · Full work rights (Subclass 485 Temporary Graduate visa)

Summary

Software engineer focused on systems and backend development. Experienced building concurrent C/C++ software involving memory management, atomics, POSIX threads and low-level debugging. Master of Computer Science from the University of Sydney and current Systems Programming tutor. Seeking a graduate or junior backend, systems or platform engineering role.

Technical Skills

Programming: C, C++, JavaScript, Python, Java, SQL, x86-64 Assembly
Systems: UNIX/C, POSIX threads, memory management, system calls, signals, IPC, Linux, GDB, Valgrind
Backend & Web: Node.js, Express.js, REST APIs, PostgreSQL, MongoDB, React.js
Tools: Git, Make, Bash

Experience

Academic Tutor – Systems Programming — Sydney, New South Wales, Australia · On-site
The University of Sydney · Part-time — *Feb 2026 – Present*

- Teach C systems programming to undergraduate students: memory layout, pointers, processes, IPC and concurrency. Run tutorials on POSIX threads, mutexes and semaphores, and help students debug memory errors using GDB and Valgrind.

Systems Engineer & Chief Technology Officer — Sydney, New South Wales, Australia · Hybrid
Alethi · Self-employed — *Jan 2026 – Aug 2026*

- Equity-based role building MCK1, a unikernel architecture that uses a high-level language and graph description to build custom kernels and runtimes. Designed the low-level system architecture for direct hardware execution, with a focus on security, performance and deterministic behaviour.

Full-Stack Engineer & Chief Technology Officer — Pune District, Maharashtra, India · Hybrid
ChessVerst · Self-employed — *Feb 2021 – Jul 2023*

- Equity-based role designing the platform architecture from wireframes through frontend, backend and database design. Built the platform using React, Next.js, PostgreSQL and REST APIs, including reusable frontend and backend components.

Associate Software Engineer — Bengaluru, Karnataka, India · Remote
Lyte Calc Technologies Pvt Ltd · Internship — *Mar 2022 – Aug 2022*

- Built an e-commerce platform using React, Express.js, Node.js and MongoDB with a reusable application architecture. Developed frontend components, backend services and REST APIs, and integrated them across the application.

Projects

MIMFS – Multithreaded In-Memory File System | **C**

- Thread-safe POSIX-style file system that works inside a memory buffer the caller provides, with no dynamic allocation. Supports directories, hard and symbolic links, sparse files and rename. Block and inode allocation is lock-free using C11 atomics, with per-inode locks rather than a single global lock, to reduce contention between concurrent operations.

LLFPL – Low-Level Floating-Point Language | **C**

- Interpreter for a deterministic expression language aimed at numerical work. Memory-maps the source and evaluates expressions directly without constructing a syntax tree, minimising intermediate memory use. Reserves all storage at start-up so evaluation performs no heap allocation, and keeps arithmetic to IEEE 754 double.

Education

The University of Sydney — Sydney, Australia
Master of Computer Science (Software Engineering) — *Feb 2024 – Dec 2025*

Nutan College of Engineering and Research (NCER) — Pune, India
Bachelor of Technology in Computer Science and Engineering — *Aug 2019 – Aug 2023*

Publication

Revolutionizing Student Engagement: Building a Cutting-Edge Interaction System. SSRN (Elsevier), 5th International Conference on Communication & Information Processing (ICCIP-2023).